AI-Assisted Software Development

Future and Next Steps

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Recent Developments

1. **Prompt cacheing**
    - Designate parts of a context as "static"
    - To avoid processing them in every query
    - Saves on token usage ie cost
    - Improves performance
2. **Claude Citation API**
    - Completion can contain reference to source documents
        - that were included into context
    - Page of a PDF document
    - Line and char reference of a text file
3. **Model Context Protocol**
    - Simplifies "Tools API" which informs LLM of available "actions"
    - Watch for emerging "MCP Server" implementations

Forthcoming Developments

1. **Tessl.ai**
    - Developing a suite of AI-powered tools
        - to support coding
        - and other areas of software development lifecycle
    - Spec-based development
2. **pragmaticcoders.com/resources/ai-developer-tools**
3. **shiftasia.com/column/top-ai-tools-for-coding-navigating-the-ai-revolution-in-software-development/**
4. **Model providers blogs**
    - Anthropic, OpenAI, Mistral, DeepSeek, ...

1. Java Babel project
    - Compiles bytecode onto CUDA / GPU
2. ELM Local Device AI (Apple M5 133TOPS)
3. Algorithmic advances (eg Deepseek OCR)
4. Better training data from our prompts??
5. Quantum Computing (in 5 to 20 years)

Beyond Transformers

Major Research Directions to overcome current limitations

1. Alternative Architectures

State Space Models (SSMs)

- Mamba, Mamba-2: Linear-time alternatives to transformers with selective state spaces
- Better at long sequences without quadratic attention cost
- Show promise for efficiency and long-context understanding

Hybrid Architectures

- Combining transformers with other mechanisms (CNNs, RNNs, SSMs)
- Example: RetNet (retentive networks) for parallel training + efficient inference

Sparse & Efficient Attention (eg BigBird)

- Longformer, BigBird, Reformer
- Reduce $O(n^2)$ attention complexity
- Enable processing of much longer contexts

Beyond Transformers...

2. Scaling Beyond Parameters

Test-Time Compute

- Models that "think longer" on harder problems
- OpenAI's o1/o3 models use extended inference-time reasoning
- Chain-of-thought, self-verification, search at inference time

Mixture of Experts (MoE)

- Activate only subset of parameters per input
- Scale model capacity without proportional compute increase
- Examples: GPT-4, Mixtral, Gemini 1.5

3. Multimodal Foundation Models

- True integration of vision, audio, text, video
- Examples: GPT-4V, Gemini, unified embedding spaces
- Move beyond "bolting together" separate encoders

4. Neurosymbolic AI

- Combining neural networks with symbolic reasoning
- Logic, knowledge graphs, formal methods
- Better abstract reasoning, planning, mathematical proof

Beyond Transformers...

5. Memory & Retrieval Systems

External Memory

- RAG (Retrieval-Augmented Generation)
- Vector databases + semantic search
- Dynamic knowledge updates without retraining

Persistent Memory

- Long-term memory systems that evolve
- Continual learning without catastrophic forgetting

Beyond Transformers...

6. New Learning Paradigms

Self-Supervised Learning 2.0

- Beyond masked language modeling
- Contrastive learning, self-distillation
- World models that predict future states

Reinforcement Learning from Human Feedback (RLHF) Evolution

- Constitutional AI, debate, recursive reward modeling
- Better alignment techniques
- Scalable oversight methods

Active Learning & Curriculum Learning

- Models that request specific training data
- Progressive difficulty in training

Beyond Transformers...

7. Biological Inspiration

Spiking Neural Networks

- Brain-like temporal dynamics
- Energy-efficient computing
- Better for neuromorphic hardware

Cognitive Architectures

- Working memory, attention mechanisms inspired by neuroscience
- Systems that learn more like humans

Beyond Transformers...

8. Algorithmic Reasoning

- Teaching models to execute algorithms
- Neural algorithmic reasoning (DeepMind's work)
- Compositional generalization
- Abstract reasoning (ARC challenge focus)

9. Hardware Co-Design

- Custom chips for AI (TPUs, Cerebras, Graphcore)
- Analog computing for neural networks
- Quantum-classical hybrid systems
- Optical computing

10. Agent Systems

- Tool use: Models calling APIs, code execution, databases
- Multi-agent collaboration: Specialized agents working together
- Embodied AI: Robots with vision-language-action models
- Autonomous planning: Long-horizon task decomposition

11. Formal Verification & Interpretability

- Mechanistic interpretability (Anthropic's research)
- Proving properties of neural networks
- Causal understanding of model behavior
- Circuit discovery in transformers

Beyond.. -

12. Energy Efficiency

- Pruning, quantization, distillation at scale
- 1-bit models (BitNet)
- Training with lower precision
- Edge deployment optimization

Near-Term Breakthroughs to Watch:

- Models with 10M+ token context (vs current ~1M)
- Real-time learning during conversation
- Reliable mathematical reasoning and formal proof
- Generalist robots powered by foundation models
- Emergent abilities from scale + novel architectures

Key Limitation Being Addressed:

Current transformers struggle with:

- Reasoning depth → Test-time compute helps
- Long-term memory → External memory systems
- Sample efficiency → Better learning paradigms
- Abstract reasoning → Neurosymbolic approaches
- Grounding → Multimodal embodied agents

Next Steps

1. **Choose Tools**
    - Experiment with trial versions
    - Map "preferred" tools to company strategy / budgets
    - Avoid proliferation of tools ie costs
    - Watch blogs.. for emerging tools
2. **Remember Data Privacy**
    - Appropriate level of caution
    - Be aware of Tool Vendor and Model Providers' policies
3. **Keep Practicing!**
    - Actively seek areas to "delegate" to AI tools
    - Look out for signs of context problems
    - Keep an eye on LLM model developents